Centre for
Data Analytics

Insight

SAFE

Policy Aware SPARQL Query Federation Over RDF Data Cubes

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Germany

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Linked2Safety - Showcases

1. *Showcase #1 – Phase III Clinical Trial: Subject Selection Criteria:*
   - the unbiased randomised selection of su[illegible]s in phase III clinical trials
   - *e.g. return subjects with diabet[illegible]d weight >80 and hasCancer*
2. *Showcase #2 – Phase IV Po[illegible]urveillance trial:*
   - *the pharmacovigil[illegible] after it receives permission to b[illegible]*
   - *e.g. Test D[illegible]ion with headaches*
3. *Showcas[illegible]nformatics:*
   - *identifi[illegible]of relations between molecular fragments and specific a[illegible]verse side effect categories.*
   - *e.g. Test chemicalFragmentX(of DrugX) with rash*

Connecting Multiple Clinical Sites
Query Federation

The Problem





return number of **patients** that have been **administered** the drug **Insulin** and exhibit **BMI** > 25 and **Hypertension** and **Diabetes** as adverse events

Switzerland

Cyprus

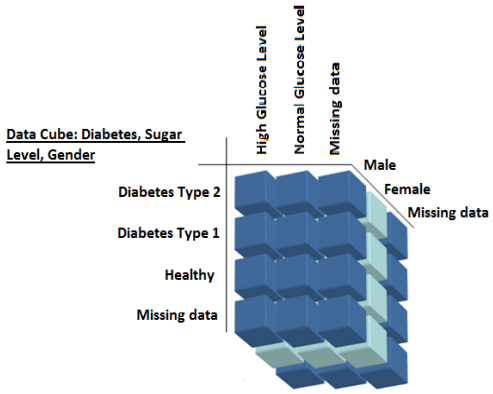
Greece

Safety First – Ethical & Legal Aspects



Patients' anonymity

- ✓ Anonymised Clinical Data cubes
- ✓ Insensitive clinical parameters without personal information



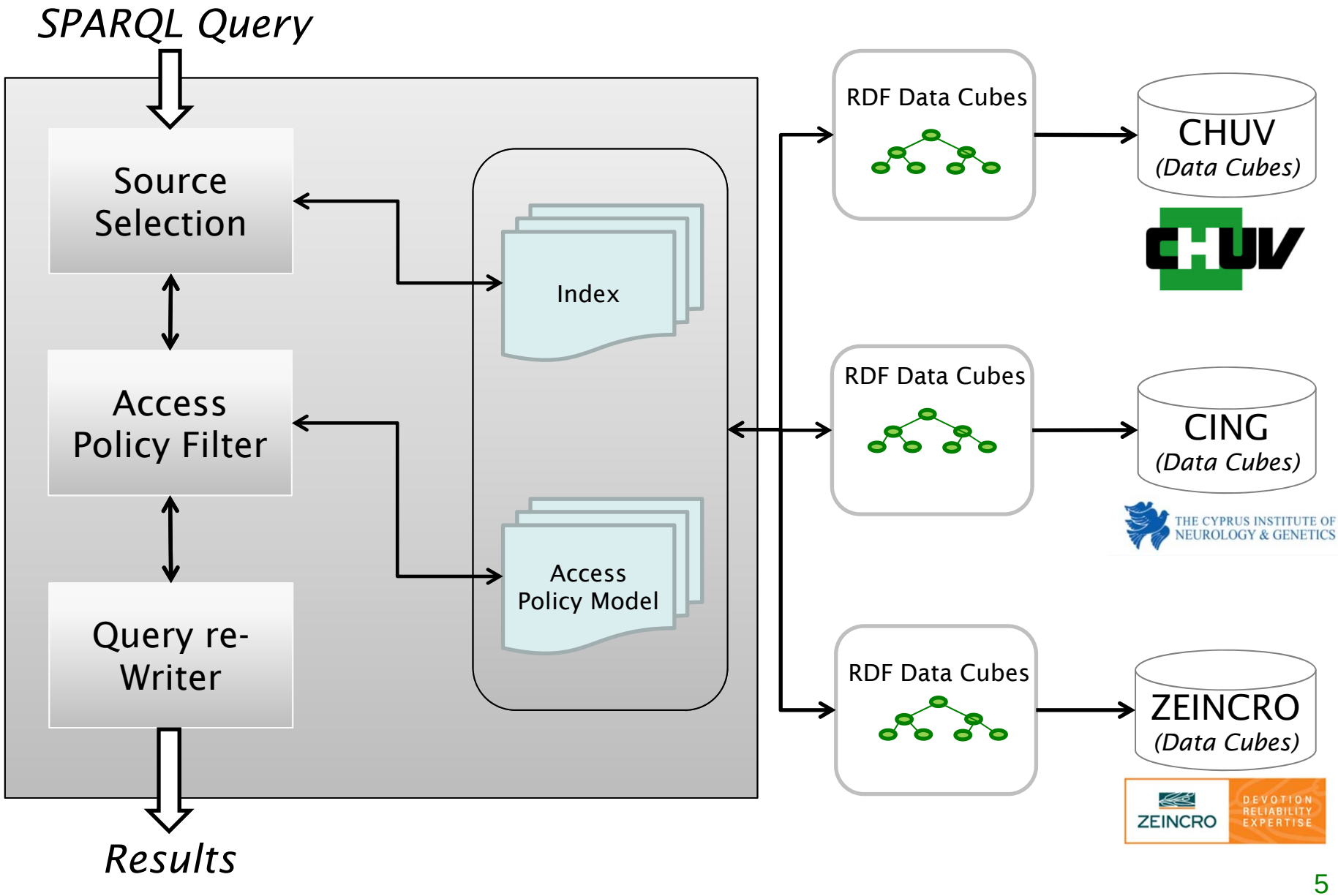


Data Ownership & Privacy

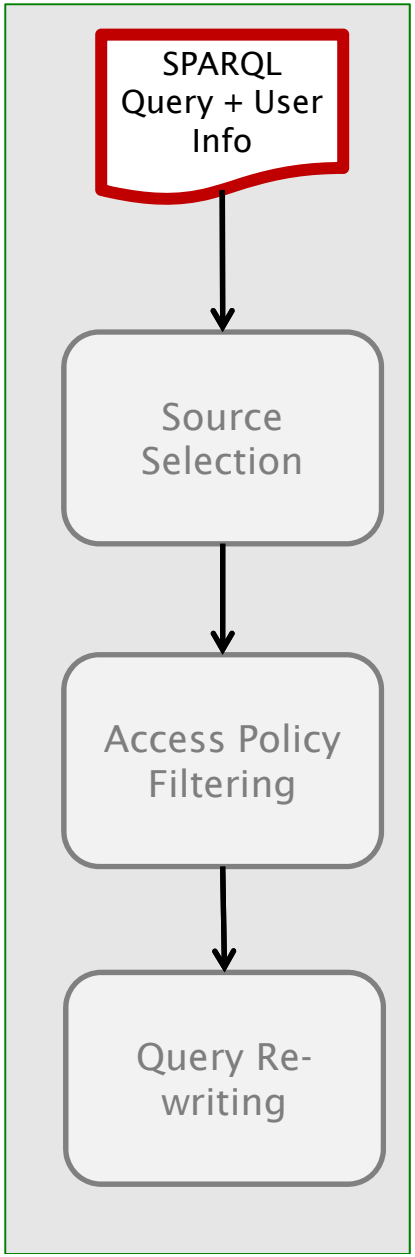
- ✓ Access-Control Based Query Federation



SAFE - Secure SPARQL Query Federation



SAFE



Oya

Clinical Researcher
Expertise – Diabetes

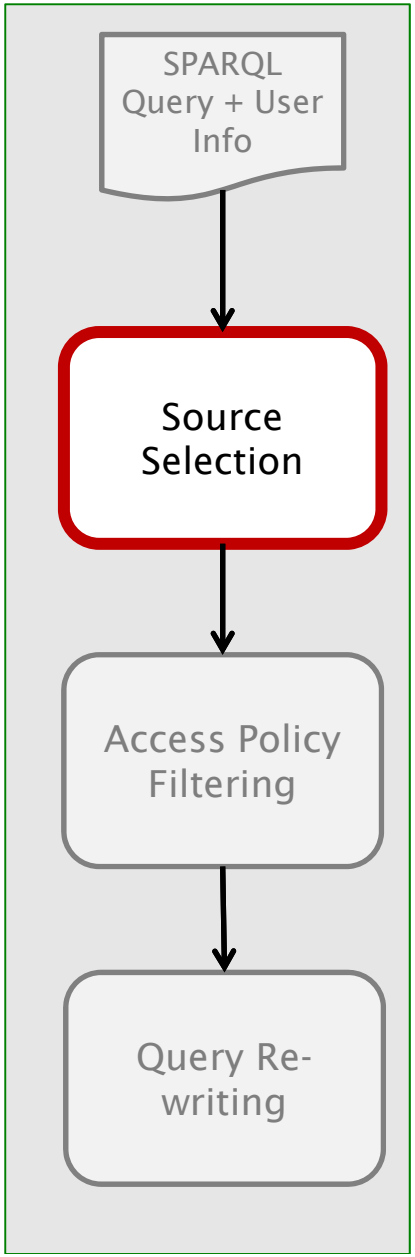


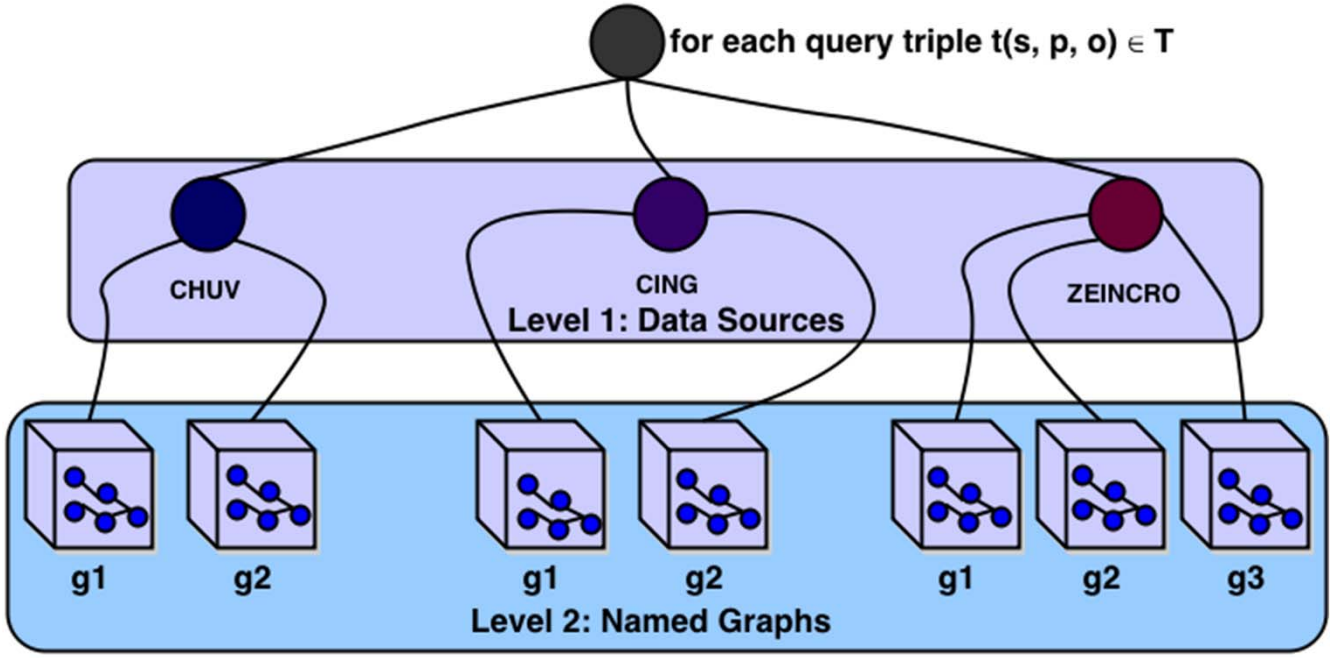
CHUV

THE CYPRUS INSTITUTE OF NEUROLOGY & GENETICS

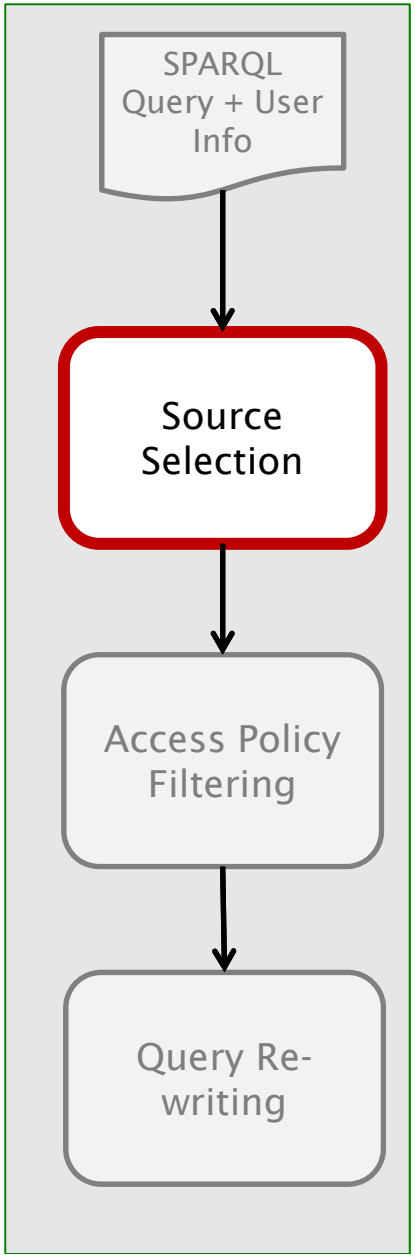
```
SELECT ?diabetes ?bmi ?hypertension ?cases
WHERE {
?observation a qb:Observation .
?observation l2s-dim:Diabetes ?diabetes.
?observation l2s-dim:BMI ?bmi.
?observation l2s-dim:Hypertension ?hypertension.
?observation sdmx-measure:Cases ?cases.
}
```

SAFE – Source Selection

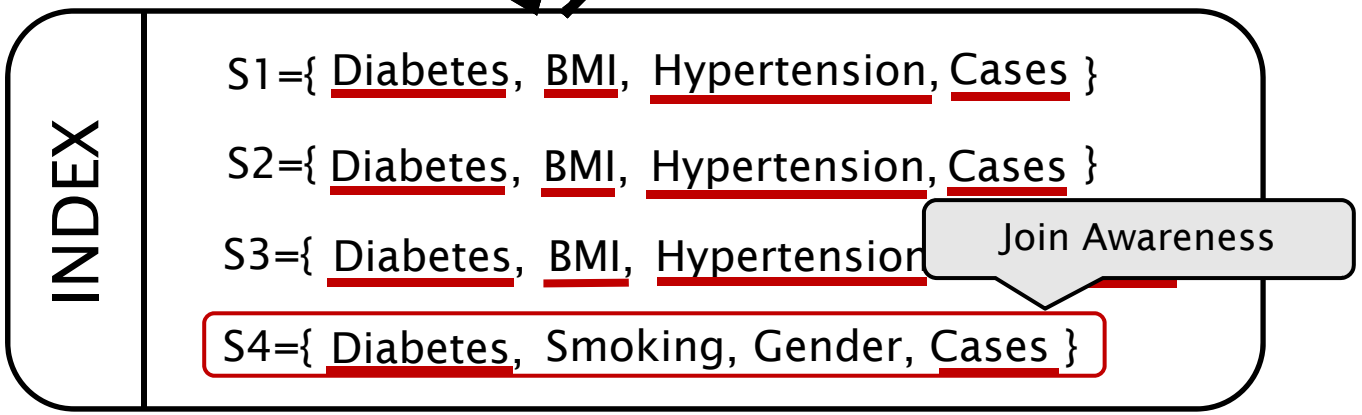




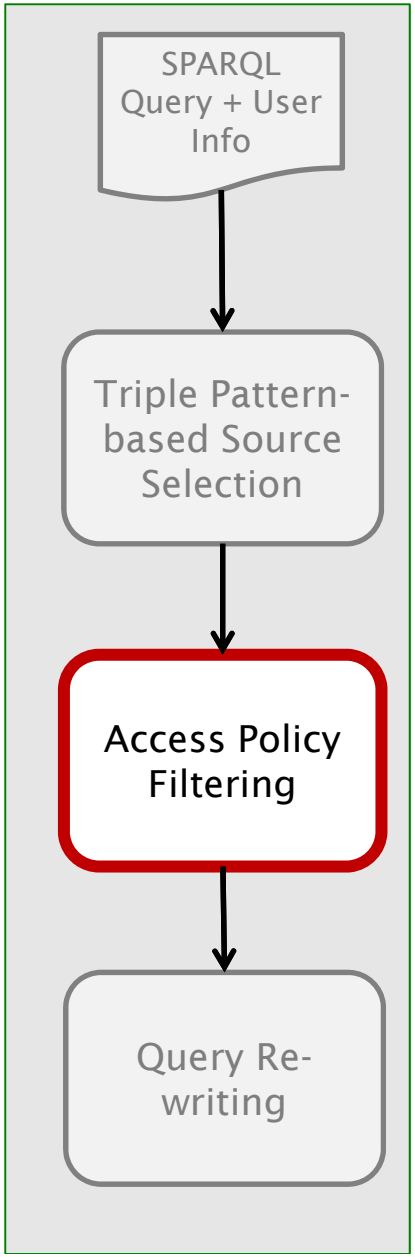
SAFE – Source Selection

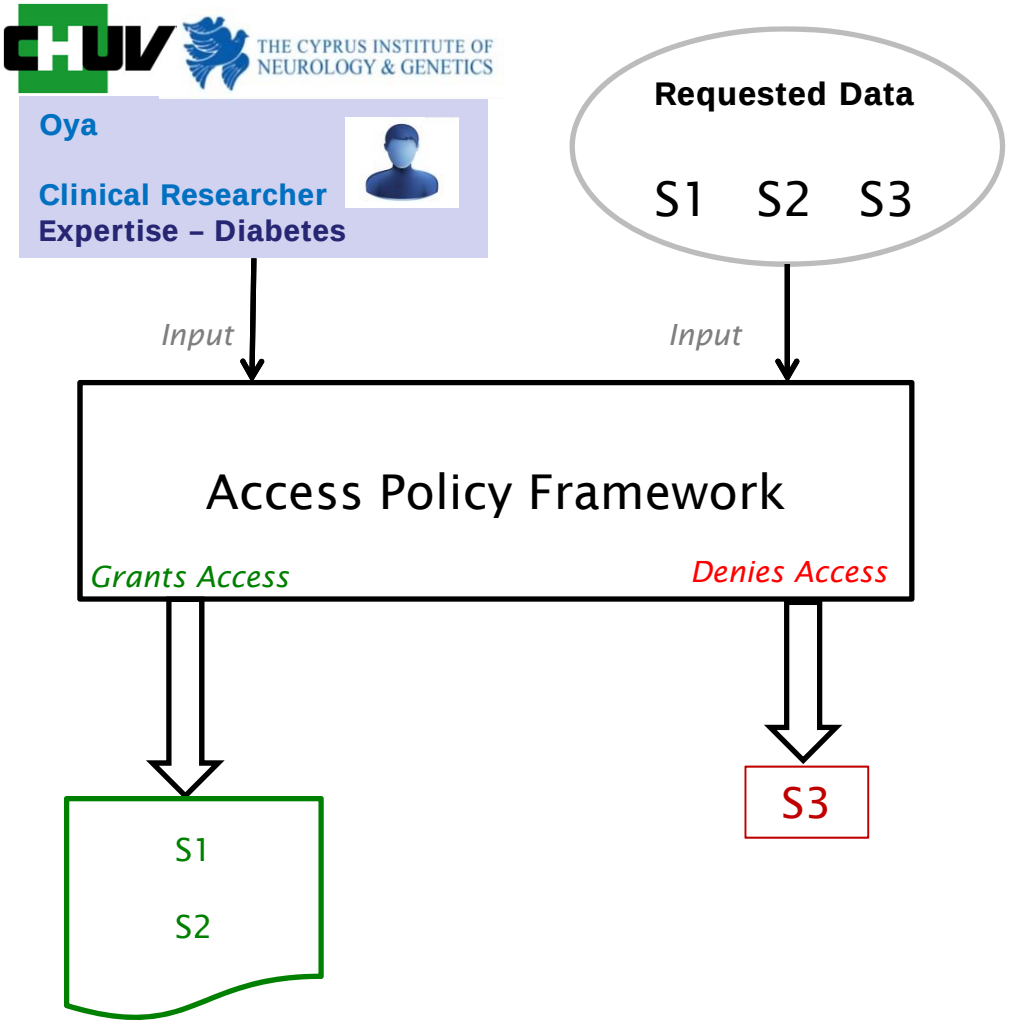


Triples Patterns	Capable Sources
?observation l2s-dim:Diabetes ?diabetes.	{S1, S2, S3, S4}
?observation l2s-dim:BMI ?bmi.	{S1, S2, S3}
?observation l2s-dim:Hypertension ?hypertension.	{S1, S2, S3}
?observation sdmx-measure:Cases ?cases.	{S1, S2, S3, S4}

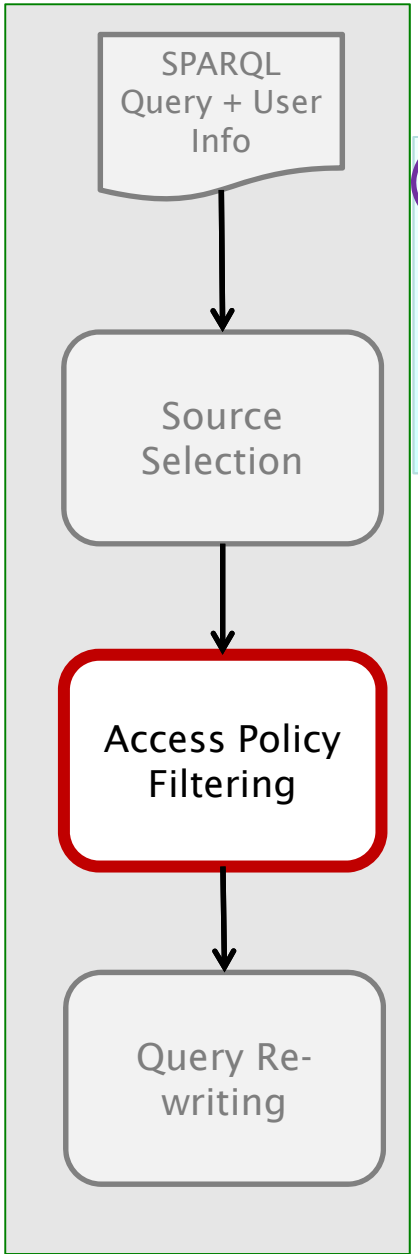


SAFE – Access Policy





SAFE – Access Policy



- Example Access Policy

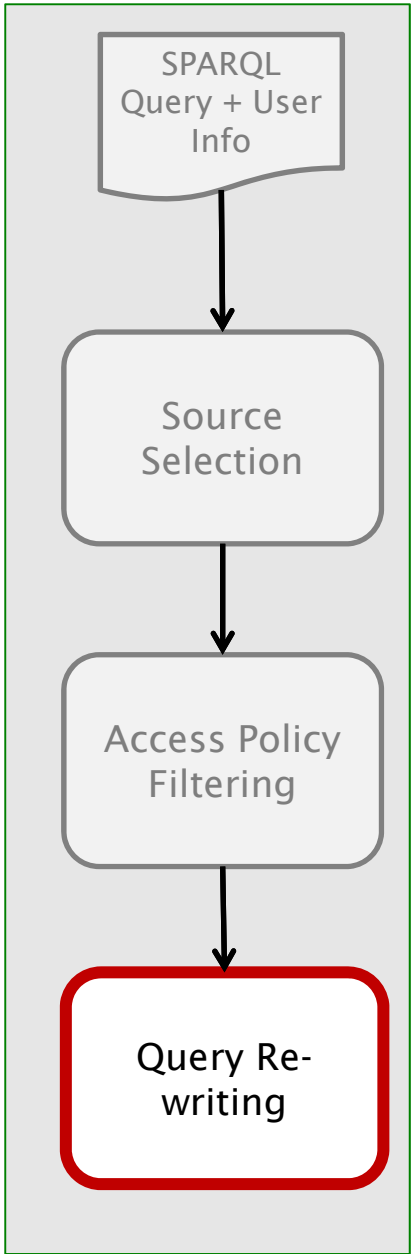
AP1	*type*	Access_Policy
AP1	*applies_to*	{S1, S2}
AP1	*grants_access*	Read
AP1	*assigned_to*	Oya

Oya	*type*	User
Oya	*haslocation*	Galway
Oya	*hasPurpose*	Perform p-value analysis
Oya	*hasRole*	Clinical Researcher
Oya	hasDomain	Diabetes

- SPARQL Query

```
ASK WHERE  {
?accessPolicy   a   AccessPolicy.
?accessPolicy   appliesToNamedGraph  S1.
?accessPolicy   grantsAccess     acl:Read_l2s,
?accessPolicy   hasUser  Oya.
}
```

SAFE – Query Rewriting



- Graph Information will be added to the query triples
  - *SELECT …. WHERE { GRAPH <S1> { …. } }*
  - *SELECT …. WHERE { GRAPH <S2> { …. } }*
- Sub queries sent to relevant sources
  - *S1*
  - *S2*
- Integration of results obtained from each sources

	Diabetes	BMI	Hypertension	Cases
S1	0	0	0	40
	1	0	1	50
S2	0	1	1	120
	1	1	1	90

Evaluation - DataSets

Dataset	# triples	# obs	# sub	# pred	# obj	# size	# index size	# index generation time
Internal Dataset								
CHUV	0.8 M	96 K	96 K	36	88	31 MB	-	-
CING	0.1 M	17 K	17 K	21	51	5 MB	-	-
ZEINCRO	0.4 M	49 K	49 K	24	59	15 MB	-	-
Total	1.3 M	162 K	162 K	81	198	51 MB	8 KB	10 sec
External Dataset								
World Bank	77 M	10 M	10 M	58	40 K	19 GB	-	-
IMF	18 M	1.8 M	1.8 M	30	3151	3.51 GB	-	-
Eurostat	0.3 M	38 K	44 K	31	5717	205 MB	-	-
Trans. Int.	43 K	3939	4286	64	5290	9.2 MB	-	-
Total	95 M	12 M	2 M	183	54 K	23 GB	12 KB	571 sec

Evaluation - Berlin SPARQL Benchmark

Characteristics	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
# of Triple Patterns	9	7	9	16	7	8	11	10	7	7	3	7
# of Sources	3	4	4	3	4	3	3	4	3	3	3	3
# of Results	41	50	348	41	62	1983	5	10	1701	19656	570	41
Filters							√					
> 9 Patterns	√		√	√	√	√	√	√				
Negation							√					
LIMIT Modifier		√	√				√	√				
Order By Modifier		√	√					√				
DISTINCT Modifier		√		√	√	√	√	√	√	√	√	√
REGEX Operator					√							
UNION Operator	√											

Evaluation – Source Selection

- Sum of triple-pattern-wise **sources selected** for each query

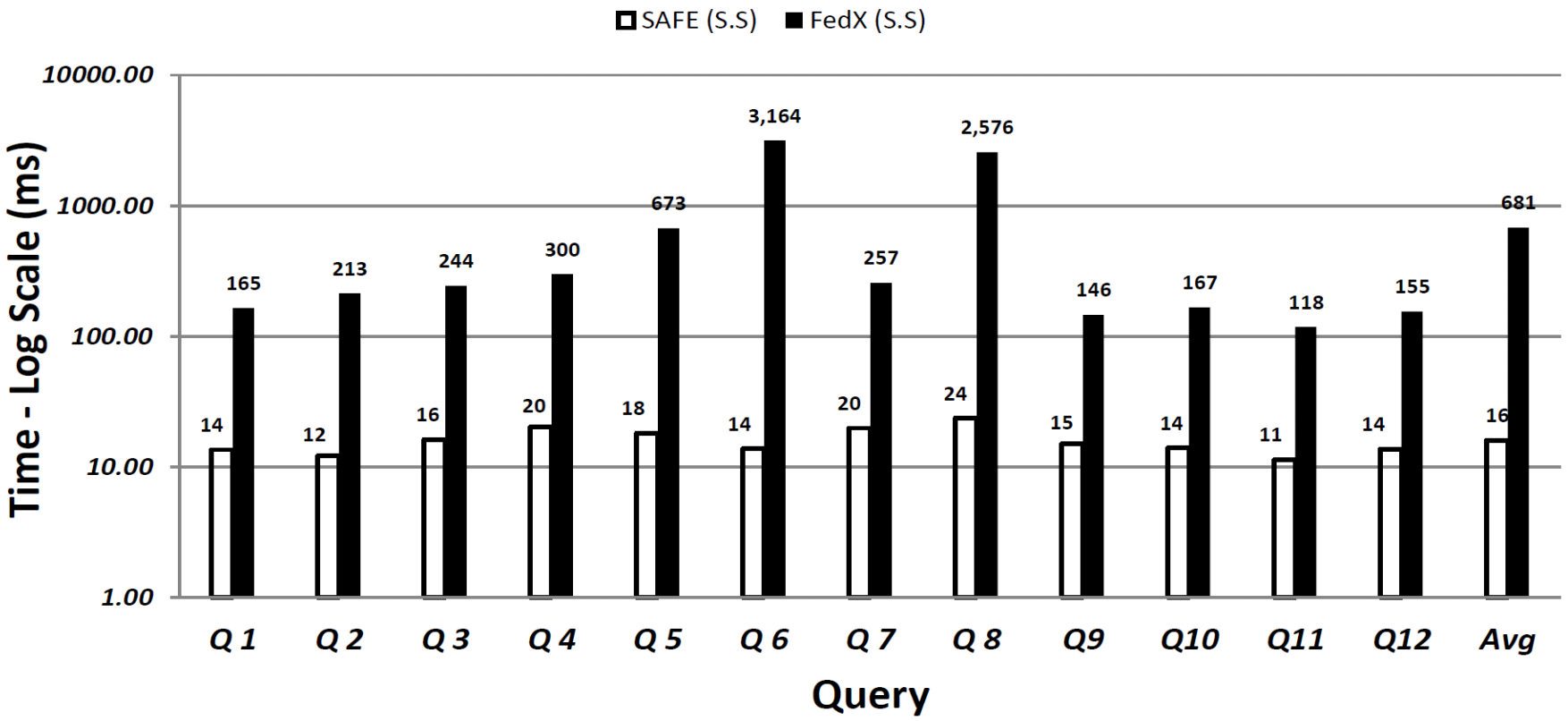
Systems	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Avg
SAFE	8	10	13	16	15	13	15	16	7	7	9	7	11
FedX	9	13	16	24	20	14	16	19	15	17	9	16	16

- Number of SPARQL **ASK requests** used for source selection

Systems	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Avg
SAFE	0	0	0	0	0	0	0	0	0	0	0	0	0
FedX	36	28	40	64	48	40	44	40	21	21	9	21	35

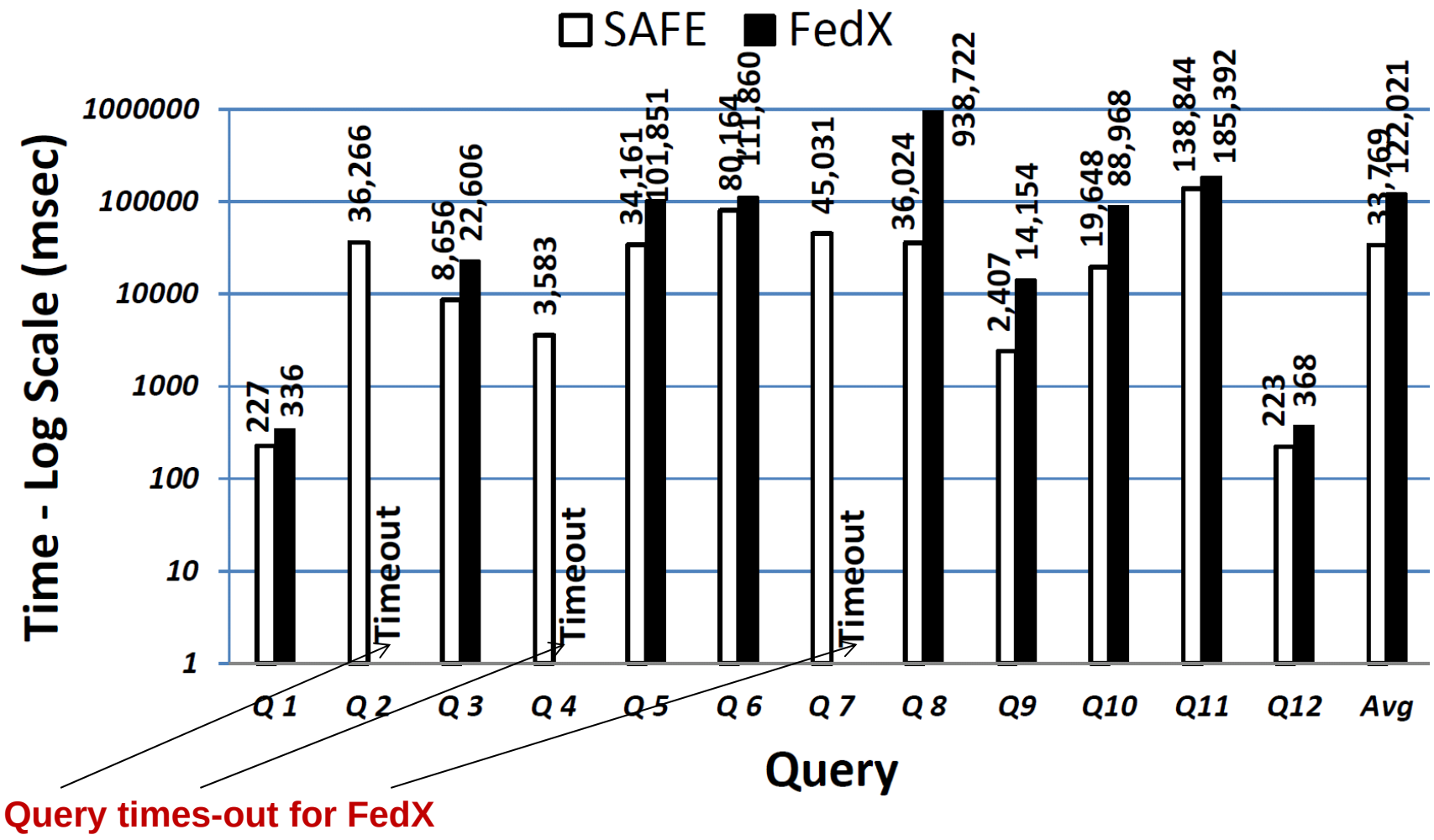
Evaluation - Source Selection Time

- **Source Selection Time**



Evaluation - Query Execution Time

- **Query Execution Time**



SAFE – Highlights

- Source Selection
  - SPARQL SERVICE
  - Using SPARQL ASK queries
  - Using a catalog/index
  - SAFE - Hybrid (catalog/index + ASK)
- Lightweight Cache
  - RDF Cube Data Structure
  - AccessPolicy
- Join Aware
  - Excludes ineligible sources before actual query join
- Provenance – via RDF Named Graphs
  - Self-contained Data Cubes
  - Creator
  - Location
  - Date
  - Access rights

Conclusion & Future Work

- ✓ Efficient **source selection** with a **lightweight indexing**
- ✓ **Policy aware** query execution
- ✓ Evaluated against **internal and external sets**
- ✓ Performance is **significantly improved** compared to FedX
- ■ Cooking at the moment !
  - □ **Evaluation extended** to federation engines (ANAPSID, HiBISCuS)
  - □ **Benchmarking** for query federation over statistical data cubes
  - □ SAFE extension for **normal RDF data**

SAFE - Team

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- Muntazir Mehdi
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- Panagiotis Hasapis
- Axel-Cyrille Ngonga Ngomo
- Stefan Decker
- Ratnesh Sahay



SAFE
POLICY AWARE SPARQL QUERY FEDERATION

http://linked2safety.hcls.deri.org:8080/SAFE-Demo/

Thank You